

Firewall, VPN or TunnelGate?

Hardware firewall vs software firewall vs VPN:
A simple comparison, with costs in Indian rupees

- Hardware firewall
- Software firewall
- VPN
- TunnelGate

THE PROBLEM IN ONE MINUTE

Your staff need to use office computers from home. The usual way is Remote Desktop (RDP). It works, but it needs a door (port 3389) open to the internet, and attackers check for that door all day, every day.

So the real question is: how do we let our own people in without leaving that door open?

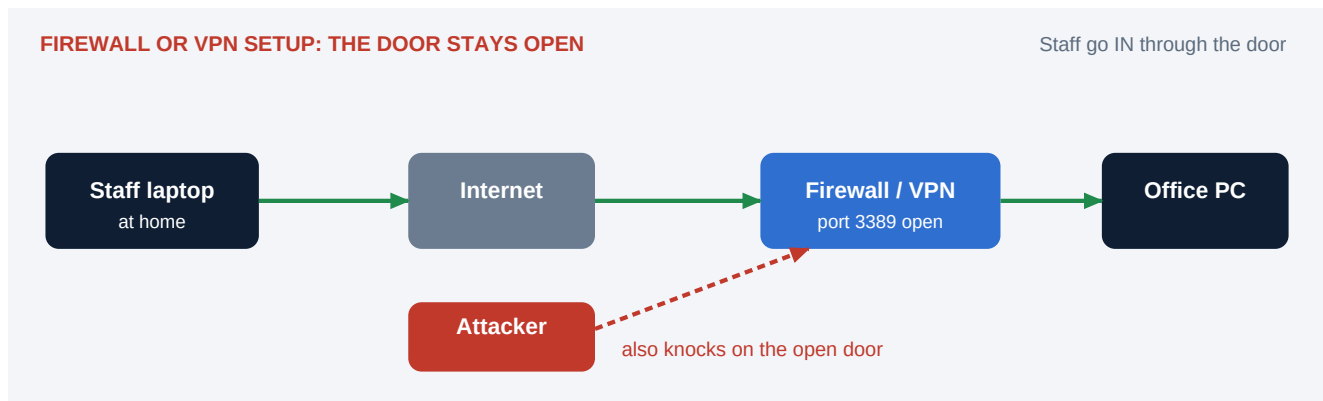
FOUR WAYS TO DO IT

Hardware firewall A security box at the office. Filters what comes in.	Software firewall A program on each PC. Filters what reaches that PC.
VPN A private tunnel from the staff laptop into the office network.	TunnelGate The PC connects out to Cloudflare. Staff click once and get in. No open door.

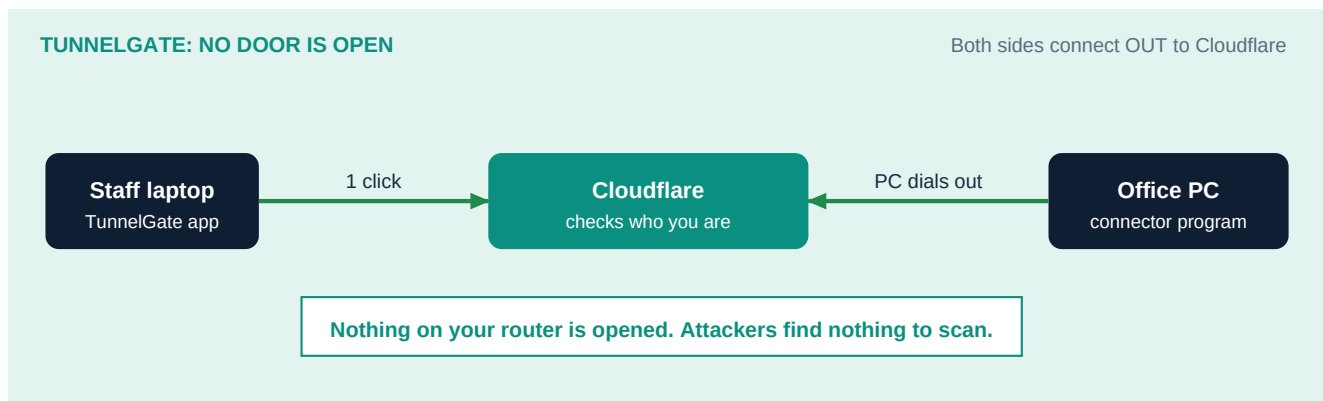
Short answer
A firewall or a VPN each fixes part of the problem, and costs time to run. TunnelGate is the only one here that keeps the door closed, lets staff in with one click, and costs the least to run. It does not replace a firewall; keep the one on each PC.

How the connection works

The difference between the options comes down to one thing: does something on your side wait for the internet to knock, or does it call out instead?



A firewall rule or a VPN server must listen for incoming connections. Staff can use that door, and so can anyone scanning the internet for it.



With TunnelGate the office PC connects out to Cloudflare, which is allowed on almost every network. Staff connect to Cloudflare too, and Cloudflare joins the two once it has checked who they are. There is no listening door to find.

This diagram shows the connection path only. Keep the firewall on each PC as well.

1 | Hardware firewall

A security box (Fortinet, Palo Alto, SonicWall and similar) sitting between your office and the internet. To allow remote desktop you either open a door for it or allow only certain addresses.

GOOD

- Protects the whole office network at once
- Strong filtering built in
- Hackers on a PC cannot easily switch it off

NOT SO GOOD

- Costs a lot: about Rs 75,000 for a small box, plus yearly fees
- Needs a network expert to set up and update
- Door stays open, or staff get locked out when their home address changes

Best fit. Larger offices that already have an IT person and a budget.

2 | Software firewall

A program on each PC, such as Windows Defender Firewall. It decides who may reach that PC.

GOOD

- Free, already built into Windows
- Protects each PC even on café Wi-Fi
- Stops one infected PC from reaching the next

NOT SO GOOD

- It only filters; it gives remote staff no way in
- Anyone with admin rights on the PC can turn it off
- Must be set up on every PC, and settings drift over time

Best fit. An extra layer on every PC. Never the only protection.

3 | VPN

A private tunnel. The staff laptop connects to a VPN server at the office, then opens remote desktop inside that tunnel.

GOOD

- Remote desktop door stays closed to the internet
- Encrypts everything, not just remote desktop
- Gives access to files and printers too

NOT SO GOOD

- You must run and update the VPN server, and VPN servers are a favourite target
- Every person needs an app, a key and set-up help
- Staff must start the VPN first; it often gets blocked on hotel Wi-Fi
- One stolen laptop can reach the whole network

Best fit. Teams that need many office systems, not only remote desktop, and have someone to run it.

4 | TunnelGate

The office PC makes an outgoing connection to Cloudflare. Staff open TunnelGate and click once. Nothing needs to be opened on your router.

GOOD

- No open door and no public address to attack
- One click for staff; works on any Wi-Fi or mobile data
- Each person only reaches the PC they are allowed to
- Cloudflare can check who is logging in and keep a record
- Free for up to 50 users, so the lowest running cost for small teams

NOT SO GOOD

- Depends on Cloudflare and on a Cloudflare account being set up
- Remote desktop only; it is not a full office network like a VPN
- Still keep a software firewall on each PC
- Above 50 users it is a paid plan (about Rs 600 per user per month)

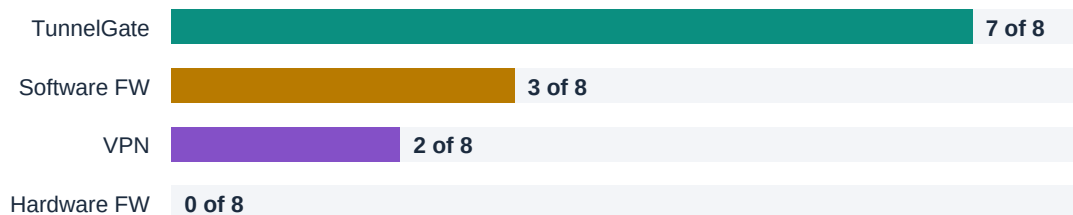
Best fit. Small and mid-size teams who want safe remote desktop without running extra equipment.

Side by side

Green is good for you, red is a problem. Ratings are my own judgement for a small team using remote desktop.

Question	Hardware firewall	Software firewall	VPN	TunnelGate
Is the server hidden from the internet?	No	No	Partly	Yes
Can staff connect from any Wi-Fi or network?	No	No	Partly	Yes
Does it give staff a way in?	Partly	No	Yes	Yes
Is login one step for staff?	Partly	Yes	No	Yes
Are staff limited to only the PC they need?	Partly	Yes	No	Yes
Is there a record of who connected?	Partly	No	Yes	Yes
Is there nothing for you to patch and run?	No	Partly	No	Partly
Is it cheap to own?	No	Yes	Partly	Yes
Number of Yes (out of 8)	0	3	2	7

How many of the eight questions get a clear Yes

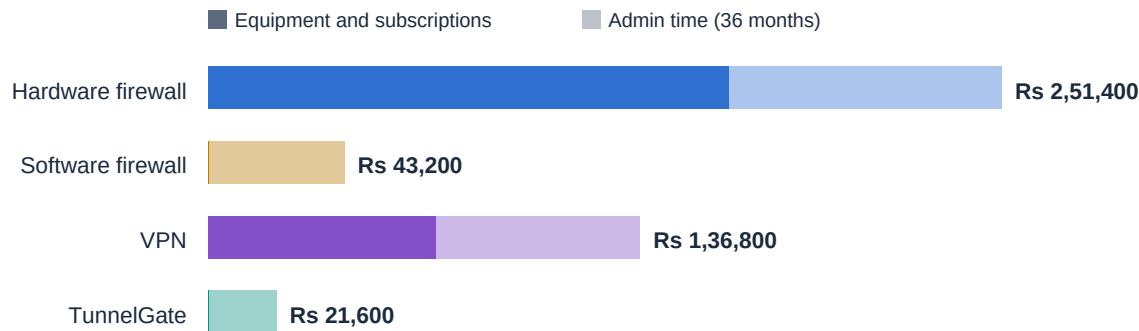


What TunnelGate does not fix

It does not replace the firewall on each PC, it relies on Cloudflare staying online, and it only covers remote desktop. That is why it scores Partly on upkeep: the small connector program on the PC still needs to run.

What it costs over three years

Example: 25 staff, 1 office, IT time valued at Rs 600 per hour. These are rough estimates in Indian rupees, not quotes.



Option	Equipment and subscriptions	Someone's time (36 months)	3-year total
Hardware firewall	Rs 1,65,000	Rs 86,400	Rs 2,51,400
Software firewall	Rs 0	Rs 43,200	Rs 43,200
VPN	Rs 72,000	Rs 64,800	Rs 1,36,800
TunnelGate	Rs 0	Rs 21,600	Rs 21,600

What is in these numbers

Hardware firewall: box about Rs 75,000, plus Rs 30,000 a year for support and updates, and about 4 hours a month of IT time. Software firewall: free, about 2 hours a month. VPN: a hosted server at about Rs 2,000 a month, and about 3 hours a month. TunnelGate: free on Cloudflare for up to 50 users, and about 1 hour a month. Above 50 users, Cloudflare charges about Rs 600 per user per month (roughly US\$7), so check their price page before you grow.

Where the money goes

For every option except TunnelGate, most of the cost is people's time: updating, fixing and setting up users, every month. TunnelGate has the smallest bill because there is almost nothing to run.

Which one should you choose?

If this is you	Choose
Small team, no IT person, want it simple and cheap	TunnelGate, plus the free firewall on each PC
Staff work from home or travel on changing networks	TunnelGate
The server must not be visible on the internet	TunnelGate or VPN
Staff need many office systems, not only remote desktop	VPN
You must protect a whole office network	Hardware firewall (and TunnelGate for remote access)

Where the facts come from

Checked 2 October 2026.

What we say	Source
Remote desktop should not be open to the internet; use zero-trust access instead	CISA BOD 23-02 and guidance
Attackers use remote desktop for ransomware	CIS summary of CISA ransomware guidance
Cloudflare Tunnel only makes outgoing connections, so no door needs opening	Connectivity options and Protect your origin server
Remote desktop can be published through Cloudflare Tunnel	RDP over Tunnel
Cloudflare records who logged in and when	Access authentication logs
Free up to 50 users, about US\$7 per user per month after that	Cloudflare Zero Trust pricing
VPN servers are often attacked	CISA Known Exploited Vulnerabilities (search for VPN products)

My own estimates, with no source: the Yes / Partly / No ratings, the hours of IT time, and the rupee prices for boxes and servers. They use about Rs 85 per US dollar. Replace them with real quotes before showing a client.